Calculus II **Integral Worksheet Answers** Name:

Evaluate the following integrals:

1. $\displaystyle\int \frac{2}{x-3}\,dx = 2\ln|x-3| + C$

2. $\displaystyle\int \frac{2}{x-3} + \frac{5}{(x-3)^2} - \frac{1}{(x-3)^3}\,dx = 2\ln|x-3| - 5(x-3)^{-1} - \frac{(x-3)^{-2}}{-2} + C$

3. $\displaystyle\int \cos^2 x\,\sin x\,dx = -\frac{\cos^3 x}{3} + C$

4. $\displaystyle\int \frac{5}{\sqrt{x^2+4}}\,dx = 5\ln\left|\frac{\sqrt{x^2+4}}{2} + \frac{x}{2}\right| + C$

5. $\int xe^{2x}\,dx = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$

6. $\int xe^{x^2}\,dx = \frac{1}{2}e^{x^2} + C$

7. $\int \sin^2(3x)\,dx = \frac{1}{2}\left(x - \frac{1}{6}\sin(6x)\right) + C$

8. $\displaystyle\int \frac{3}{x^2+4x}\,dx = \frac{3}{4}\ln|x| + \frac{-3}{4}\ln|x+4| + C$

9. $\displaystyle\int 2x\cosh(3x)\,dx = \frac{2}{3}x\sinh(3x) - \frac{2}{9}\cosh(3x) + C$

10. $\displaystyle\int \sqrt{5-x^2}\,dx = \frac{5}{2}\left(\sin^{-1}\left(\frac{x}{\sqrt{5}}\right) + \frac{x}{\sqrt{5}}\,\frac{\sqrt{5-x^2}}{\sqrt{5}}\right) + C$

11. $\displaystyle\int \frac{x}{x^2-25}\,dx = \frac{1}{2}\ln|x^2-25| + C$

12. $\displaystyle\int e^{2x}\sinh(5x)\,dx = \frac{25}{21}\left(\frac{1}{5}e^{2x}\cosh(5x) - \frac{2}{25}e^{2x}\sinh(5x)\right) + C$

13. $\displaystyle\int \frac{2x}{x^2+6x+5}\,dx = \int \frac{5}{2(x+5)} - \frac{1}{2(x+1)}\,dx = \frac{5}{2}\ln|x+5| - \frac{1}{2}\ln|x+1| + C$

14. $\displaystyle\int \frac{x-1}{x^2+6x+9}\,dx = 4(x+3)^{-1} + \ln|x+3| + C$

15. $\displaystyle\int \frac{1}{x^2+6x+14}\,dx = \frac{1}{\sqrt{5}}\tan^{-1}\left(\frac{x+3}{\sqrt{5}}\right) + C$

16. $\displaystyle\int \frac{1}{\sqrt{x^2+6x+14}}\,dx = \ln\left|\frac{\sqrt{5+(x+3)^2}}{\sqrt{5}} + \frac{x+3}{\sqrt{5}}\right| + C$

17. $\displaystyle\int \frac{2x+3}{x^2+6x+14}\,dx = \ln|(x+3)^2+5| - \frac{3}{\sqrt{5}}\tan^{-1}\left(\frac{x+3}{\sqrt{5}}\right) + C$

18. $\displaystyle\int \frac{3\sin x\,dx}{\cos^2 x+9} = -\tan^{-1}\left(\frac{\cos x}{3}\right) + C$

19. $\displaystyle\int x^4\sin(2x)\,dx = -\frac{1}{2}x^4\cos(2x) + x^3\sin(2x) + \frac{12}{8}x^2\cos(2x) - \frac{24}{16}x\sin(2x) - \frac{24}{32}\cos(2x) + C$

20. $\displaystyle\int \frac{\sin(\ln 3x)}{2x}\,dx = -\frac{1}{2}\cos(\ln(3x)) + C$

21. $\displaystyle\int \frac{4x+2}{x^2-x}\,dx = 6\ln|x-1| - 2\ln|x| + C$

22. $\displaystyle\int \frac{6x^2-7x}{3x-2}\,dx = x^2 - x - \frac{2}{3}\ln|3x-2| + C$